Software Testing and Validation

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Corso di Laurea in Informatica

Testing Preliminaries

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From Formal Verification to Testing

- Main analogies: both formal verification and testing are about checking some properties of a system
  - easiest property: does the system output the correct answer for any given input?
  - other properties: does it deadlock? does it run within given deadlines? does it crash?
- Main difference: formal verification requires a formal *model* of the system and a specification of the properties in some temporal logic
  - in some cases, the model can be automatically built (e.g., for hardware verification or bounded software verification)
- Testing requires the current version (for part) of the *actual* software
  - as for the property, no need that any temporal logic is used, though it may help
  - a simulator may be used for some physical components

From Formal Verification to Testing

- Testing requires the current version (for part) of the *actual* software
  - i.e., do not modify the code before testing, unless those changes are kept in the final code
  - i.e., if you modify the software, test it, and obtain PASS, you cannot say the PASS holds also for the original software
  - in the following, it may be the case that the software is modified: all modifications are meant to be kept
    - possible modifications: code instrumentation, adding of assertions
    - possible exception: code instrumentation

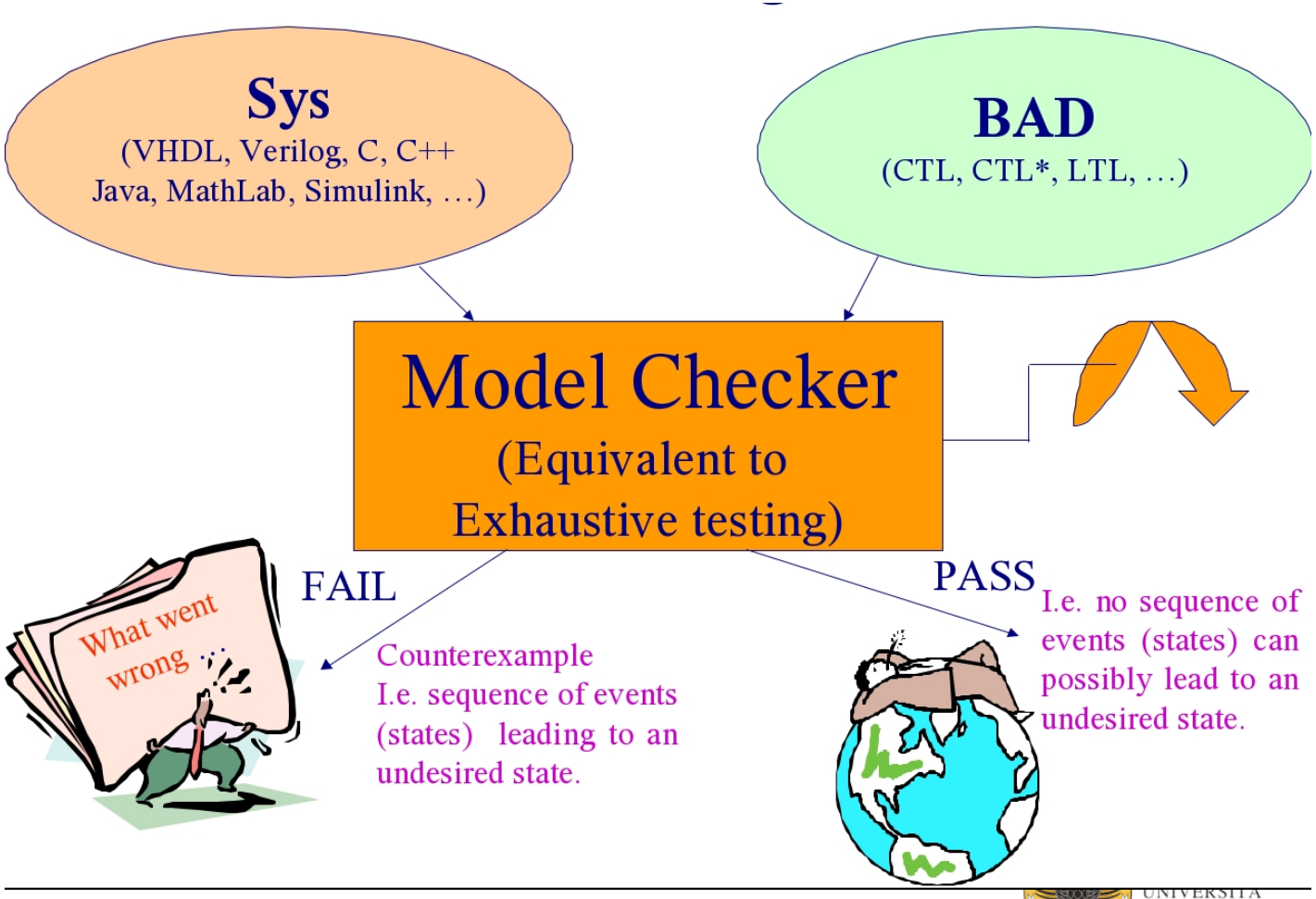
From Formal Verification to Testing

- Thus, testing is typically applied *late* in the design process
  - you need actual software, which is typically developed after architectural design and so on
  - at least for complex software projects
- However, if the software design process is well organized, testing may also be applied much early
  - e.g.: some components may be fully developed before others
  - as soon as they are developed, they may be tested
  - this is actually what it should be *always* done
  - the technique allowing this is called *scaffolding*

From Formal Verification to Testing

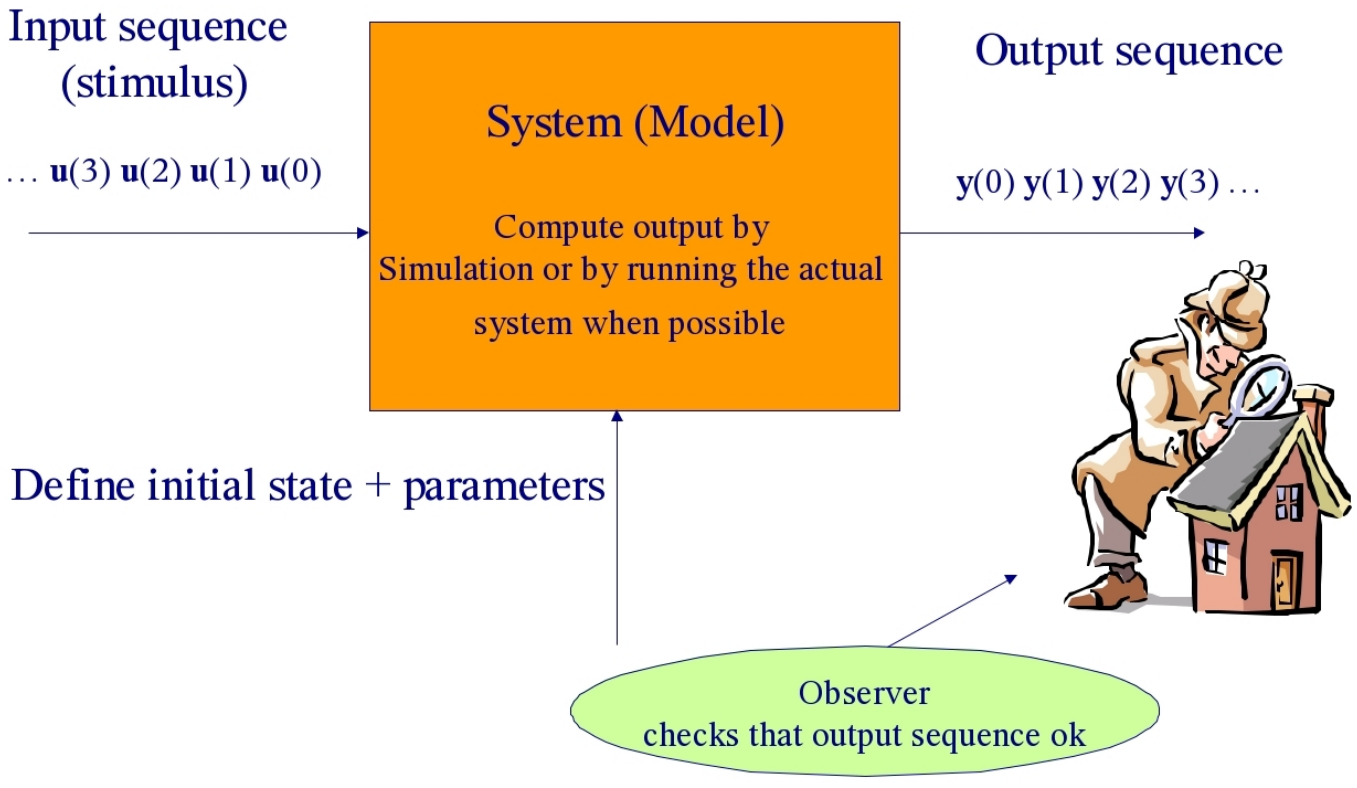
- So, no models in testing? NO!
  - you may not have a model of the system itself, but models however play an important role
  - in some cases, also a model of the system is available, why not to use it?
- Models in testing are typically used:
  - to generate inputs
  - to guide in generating inputs
  - to undestand if a testing phase is "adequate" or not
- What about algorithms?
  - no "real" algorithms are used in testing
  - forget μ-calculus or nested DFS or so on
  - though, as we will see, some algorithms may be helpful, exactly as for the models

From Model Checking...



... to Testing

An approximate answer
BUG HUNTING: Testing + Simulation



Basic Notions on Testing

- No need of complex algorithms as in model checking: simply
  1. choose what to test
     - SUV: System under Verification
  2. devise some relevant inputs
  3. execute the SUV with such inputs
  4. check if the corresponding results are ok or not
- Some automatization may be performed, but mainly a manual work for each of such steps
- Does this mean testing is easy? Obviously, NO!

Basic Notions on Testing

- Main difficulties for "choose what to test"
  - only the whole system?
    - for cyber-physical systems, a simulator may be used
  - all/some meaningful parts?
  - all/some functionalities?
  - all/some single functions or classes?

Basic Notions on Testing

- Main difficulties for "devise some relevant inputs"
  - are there some "hidden" inputs?
    - e.g., a global/class member variable, a database or a RESTful querying system
  - which are the input variables domains?
  - which values have to be selected?
    - variable domains may be extremely large (integers) or close to infinite (floats, strings)
    - sometimes, also go outside variables domains
  - once values are selected, may we understand which share of the input space got covered?
  - all inputs at the beginning, or must be fed with some timing?

Basic Notions on Testing

- Main difficulties for "execute the SUV with such inputs"
  - whole system: sometimes easy, sometimes hard
    - a simple Web interface and a submit button: easy
    - sequence of commands in CLI
    - only invokable by some mouse clicks sequence (hidden functionality)
    - many library requirements
    - ...
  - only some part(s): must be extracted in some way for execution
    - all dependencies must be resolved
  - all inputs at the beginning, or must be fed with some timing?

Basic Notions on Testing

- Main difficulties for "check if the corresponding results are ok or not"
  - in many cases (usually, not for the whole system...), the result isn't easy to extract
    - it may be in some global variable, needs some extra code to output its value after the function call
    - or in a private class member without a getter: the getter must be added (or the private removed)
    - the result may be put in some file/database/socket
  - but also after we got the result: how to know if it is correct? isn't it a circular reasoning?
  - if we simply want to detect failures, it may be needed to modify the code to expose them
    - segmentation faults for C/C++ code...
  - check for software usability: what is "ok" is not well defined...

Basic Notions on Testing

- No general tool is available for none of the problems above
  - some subparts may be automatized for many interesting cases, but not for all cases
- Testing must be integrated within software process
  - only testing at the end of the developing phase is bad
- Running tests has a cost: consider project budget and release deadlines
  - may force to only check the whole system with very limited time

Basic Notions on Testing

- Model Checking is only performed for mission- or safety-critical systems with medium-high budget
- Testing is *always* performed on *any* software
  - from cli-based computer-science-first-year projects to airport management system
- Even more: any released version of a software must be tested
  - otherwise, CrowdStrike-like problems
  - easy to understimate the effect of a modification on a working software
  - Regression Testing

Basic Notions on Testing

- Testing all features is typically impossible for complex projects
- For simpler projects, all features may be tested, but not at 100% coverage
  - actually, also 1% testing coverage is often too difficult to obtain
  - even if there are some particular measures for which 100% coverage may be achieved
- Priorities must be defined
  - more risky and defect-likely parts of the software must be checked first
  - strictly followed by the parts which are executed more often
  - 80/20 rule: 80% of any daily business activity is provided by 20% of the business system functions, transactions, or workflow

Basic Notions on Testing

- Some authors distinguish between *positive* vs *negative* testing
  - positive testing: check that the software behaves as it should under "normal" conditions
  - negative testing: check that users cannot "break" the system
    - both if they are malicious: denial of service and similar issues
    - and if they are not: including apex by accident in a non-sanitized search...
- We will consider both these types of testing without further distinction

Basic Notions on Testing

- Software related figures:
  - architects: general software "structure"
  - developers: write (and fix!) code
  - testers: plan, prepare and execute tests; furthermore, interpret test results
- For not-too-big projects, some or all of these figures could coincide
- For complex projects, testers should be separated from developers and/or architects
  - though they must know the corresponding skills
  - not influenced by having devised and/or implemented the system
  - the same programmer could be a developer in a project and a tester in another

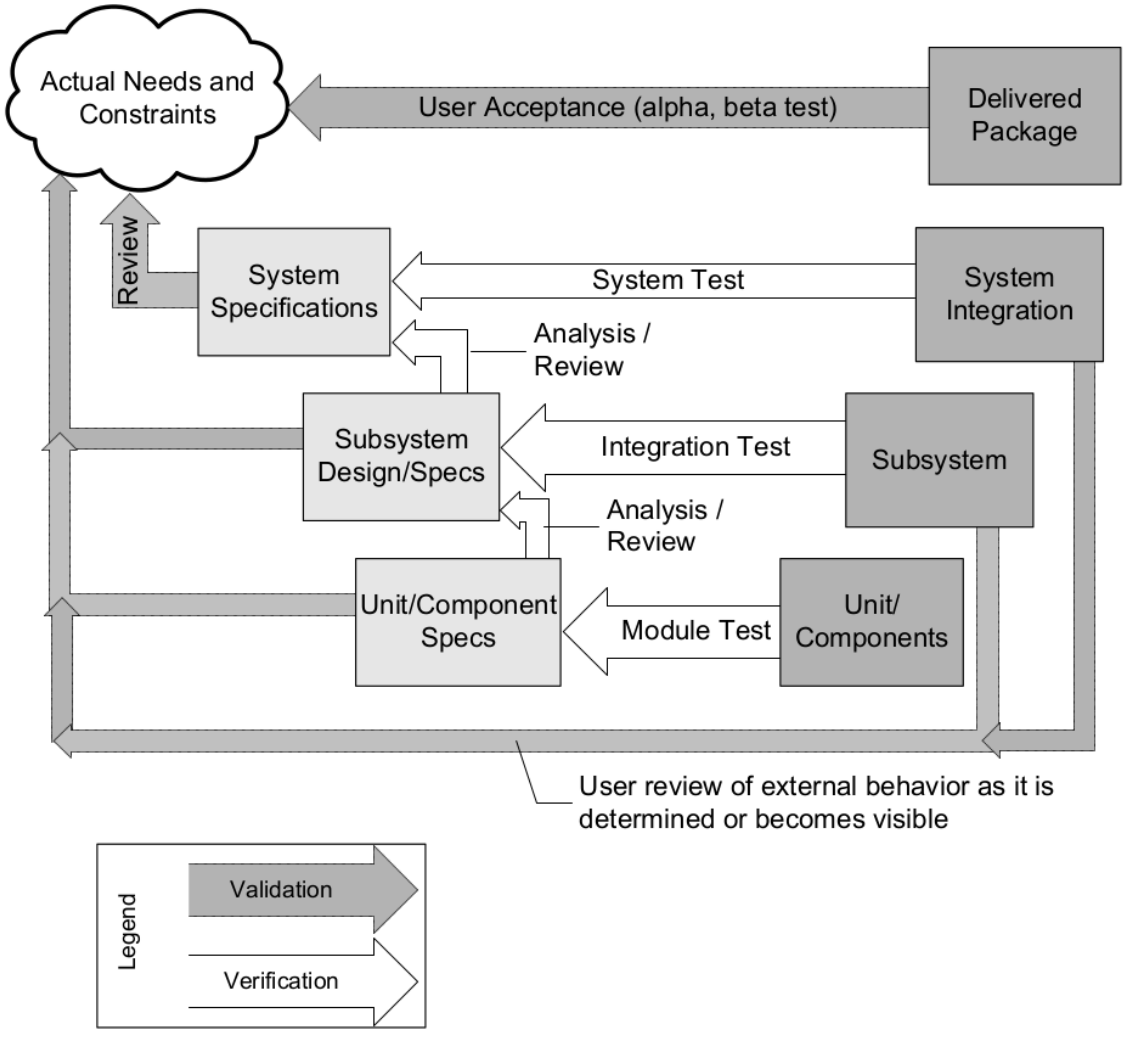
Basic Notions on Testing

- Skills of a software engineer specialized in testing: matches the steps from above
  - choose what to test
  - devise some relevant inputs
  - execute the SUV with such inputs
  - check if the corresponding results are ok or not
- All keeping into account budget (i.e., number of testering engineers) and deadlines (i.e., time available for testing)
- In a testing team, such capabilities could be properly mixed

Testing Timeline

- Let us consider complex projects: the following types of testing can be performed
  1. *unit testing*: test simple functions/classes/processes first
  2. *integration testing*: put some meaningful subsets of functions/classes/processes together and test them
  3. *system testing*: test the whole system
     - last step of integration...
  4. *acceptance testing (validation)*: test the whole system with the final users
  5. *regression testing*: how to re-test the system when new releases are issued
     - code (and possibly specifications) is modified

Testing Timeline



Testing Timeline

- Some of these steps may be deleted
  - for cli-based computer-science-first-year projects, unit testing coincide with system testing
  - for medium-size projects, integration testing and system testing may coincide
  - for a library, system testing may not exist
  - for a personal software, validation is straightforward as developers and final users coincide
- Not necessarily in cascade
  - errors discovered in later steps typically cause earlier steps to be re-run
  - sometimes not only re-running, but also devising new inputs could be required
- If errors are discovered, developers have to fix them; then, re-run testing

Testing Timeline

- It is easy to understand why you need regression testing
- Why not directly acceptance testing?
  - going to the final user with a non-working project is obviously a bad idea
  - without testing, your software may simply don't run at all
- Why not directly system testing?
  - same as asking "why do you use software engineering techniques to write complex software?"
  - testing is bottom-up (from units to system) because:
    - it is far easier for developers to understand where errors are
    - allows reusability of units

Testing Main Techniques

- Two main overall methodologies:
  - *functional testing*: tester knows specs but not the code
    - also known as *black-box testing*
  - *structural testing*: tester exploits code knowledge
    - also known as *white-box testing* and *glass-box testing*
    - includes *data-flow testing*
- Two other methodologies
  - *static testing*: code may not exist, look at documentation only
    - mainly done through model checking
    - but also prototyping may be used
  - *performance testing*: needs actual code
- Applicable to all types of testing, from unit to acceptance

Testing Main Techniques

- Orthogonal and/or auxiliary techniques:
  - *combinatorial testing*
    - given some values for single inputs, obtain a full input
  - *model-based testing*
    - extract inputs from models of software
    - special case: *fault-based testing*
  - *test execution*: not always straightforward
- Applicable to nearly all testing of the previous slide

Basic Notions on Testing

- Testing is not only for software: nearly all products must be tested before being sold
  - i.e., stressed in a controlled environment
- Typically, the testing phase is standardized for a given product
  - always repeated for some randomly chosen instance of the product
  - e.g., take a smartphone from a selling pack and drop it from 10m
- For products which are not built in series, testing must be individual
  - race cars, houses, etc.
- Of course, some guidelines may be available
  - e.g., testing of houses in a seismic environment

Basic Notions on Software Testing

- Software is among the most difficult things to be checked
  - it is virtually always "customized", thus each software needs its own testing phase
- There are guidelines, some of which will be covered in this course
- Some difficulties:
  - only errors presence can be proved
  - cost
    - it is easy to make some simple tests
    - it may be enough for very-non-critical software
    - for most software, a tradeoff is needed between testing cost and software criticality

Basic Notions on Software Testing

- Some difficulties (continued):
  - non-linearity
    - if you successfully test an elevator to be able to carry 1000 kg, then it will be ok with 900 kg or less
    - if you successfully test a sorting procedure with 1000 elements, it may fail with 2 elements
    - if you make a small modification to a pair of glasses, you do not need to run full design test from scratch
    - if you make a small modification to a software (e.g., a security update), it may cause some failure in other previously tested parts of the software
    - recall the CrowdStrike vulnerability...

Six Principles for Testing (and Verification)

- The following principles characterize the Testing and Validation as an activity of its own:
  - partition, visibility, feedback
    - nothing new: also other engineering activities may use this
  - sensitivity, redundancy, restriction
    - specific for testing and validation

Six Principles for Testing (and Verification)

- *Sensitivity*
  - problem: many errors may not be "observable"
  - e.g., a buffer overflow in C/C++ may or may not cause a failure in the running process
  - sensitivity asks that errors or faults in the software always result in observable failures
  - especially hits in code design/implementation: add assertions or similar code fragments
    - or use languages with dynamic checks such that Java, Python or Rust
  - as for verification, model checking is actually more suited for sensitivity
  - also manual code inspection may be used

Sensitivity Example

```
1  /**
2  * Worse than broken: Are you feeling lucky?
3  */
4
5  #include <assert.h>
6
7     char before[ ] = "=Before=";
8     char middle[ ] = "Middle";
9     char after[ ] = "=After=";
10
11 void  show() {
12    printf("%s\n%s\n%s\n", before, middle, after);
13 }
14
15 void  stringCopy(char *target, const char *source, int howBig);
16
17 int  main(int argc, char *argv) {
18    show();
19    strcpy(middle, "Muddled");      /* Fault, but may not fail */
20    show();
21    strncpy(middle, "Muddled", sizeof(middle)); /* Fault, may not fail */
22    show();
23    stringCopy(middle, "Muddled",sizeof(middle)); /* Guaranteed to fail */
24    show();
25 }
26
27 /* Sensitive version of strncpy; can be counted on to fail
28 * in an observable way EVERY time the source is too large
29 * for the target, unlike the standard strncpy or strcpy.
30 */
31 void  stringCopy(char *target, const char *source, int howBig) {
32    assert(strlen(source) < howBig);
33    strcpy(target, source);
34 }
```



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Six Principles for Testing (and Verification)

- *Redundancy*
  - in a broad sense: having some behavior that depend on something other
  - you declare an ‘intent”, so we can test if the intent is fulfilled
  - typed languages are a type of redundancy by intent
    - e.g., you declare something to be integer and you can raise an error if instead there is a float
  - as for actual testing: check if an implementation is ok w.r.t. its specification is actually a type of redundancy
  - specifications should be written so as to ease automatic testing or manual inspection

Six Principles for Testing (and Verification)

- *Restriction*
  - your desired property is too difficult to attain?
  - restrict it, i.e., try with something easier
    - but however meaningful
  - e.g.: too difficult to check that a variable is always initialized, then check if there exist the possibility that it may be uninitialized

Restriction Example

```
1       /** A trivial method with a potentially uninitialized variable.
2         * Maybe someCondition(0) is always true, and therefore k is
3         * always initialized before use ... but it's impossible, in
4         * general, to know for sure. Java rejects the method.
5         */
6       static void questionable() {
7           int k;
8           for (int i=0; i < 10; ++i) {
9               if (someCondition(i)) {
10                  k = 0;
11              } else {
12                  k += i;
13              }
14          }
15          System.out.println(k);
16      }
17  }
```

Six Principles for Testing (and Verification)

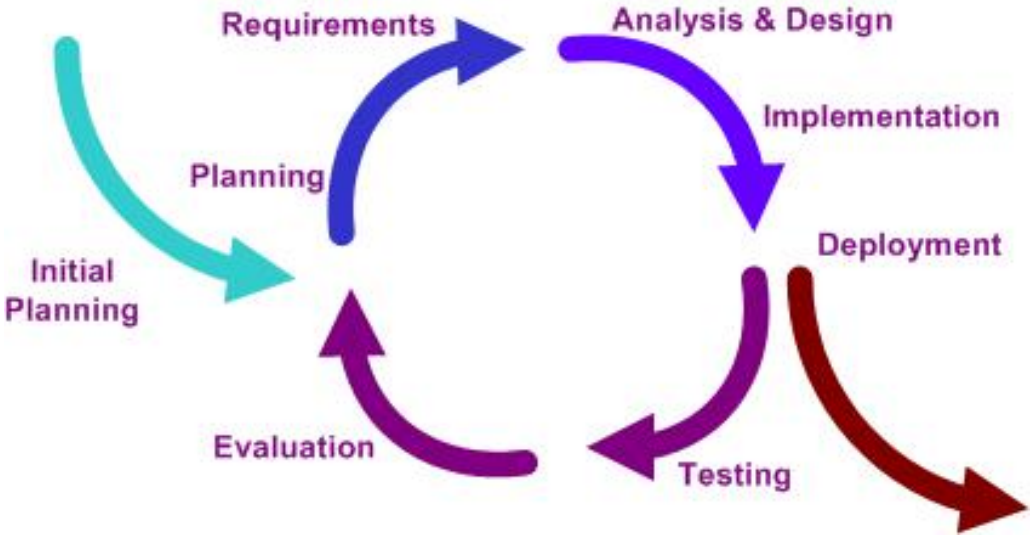
- *Partition*
  - divide and conquer (divide et impera)
  - decompose the problem to be tested
  - the very fact that many different testing techniques exists, and may be employed on the same software, it is a matter of partition
    - unit testing, functional testing, structural testing...
  - also making a model of the system is a partitioning technique
    - from "does this software satisfy my property?"...
    - to "does this model satisfy my property?" and "does this model faithfully represent the software?"

Six Principles for Testing (and Verification)

- *Visibility*
  - very similar to sensitivity, but with focus on input rather than output
  - again, mainly a design issue to ease testing
  - typical example: base program information on textual files rather than binary files
    - low performance degradation, but much better readability and capability of testing
  - e.g., HTTP exchange information as text
  - e.g., Unix-based OSs use text files for configuration
- *Feedback*
  - learn to build better testing phase from previous testing phase

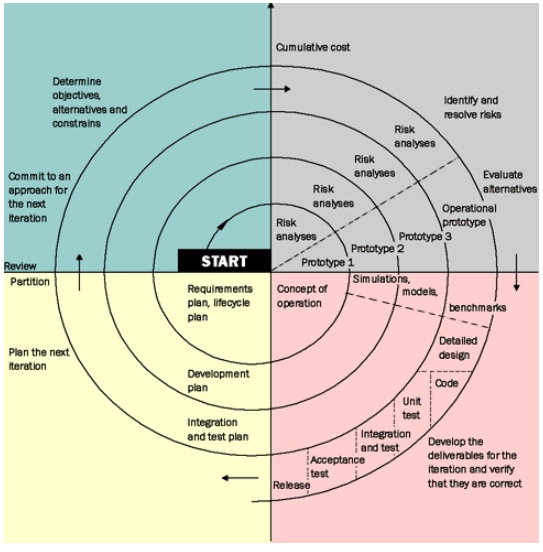
Software Process

- Not "process" in the sense of operating systems: "software process" is the whole set of activities needed to develop a high-quality software for some specific problem
  - software process contains: requirement analysis and specification, software design, implementation, validation and verification
  - organized in many ways
- Testing (and verification in general) cannot be simply done at the end



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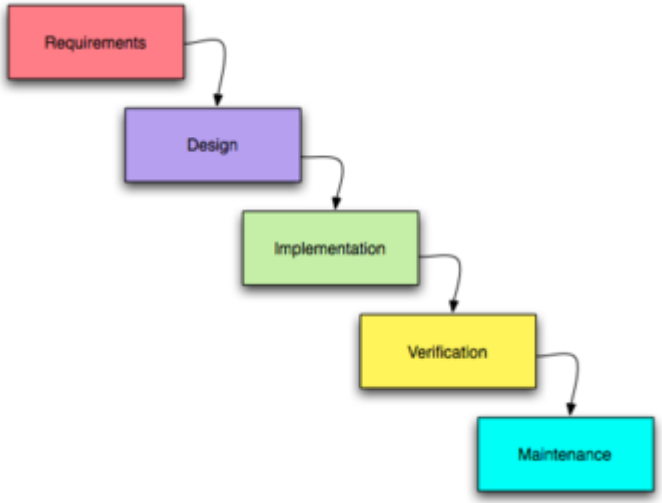


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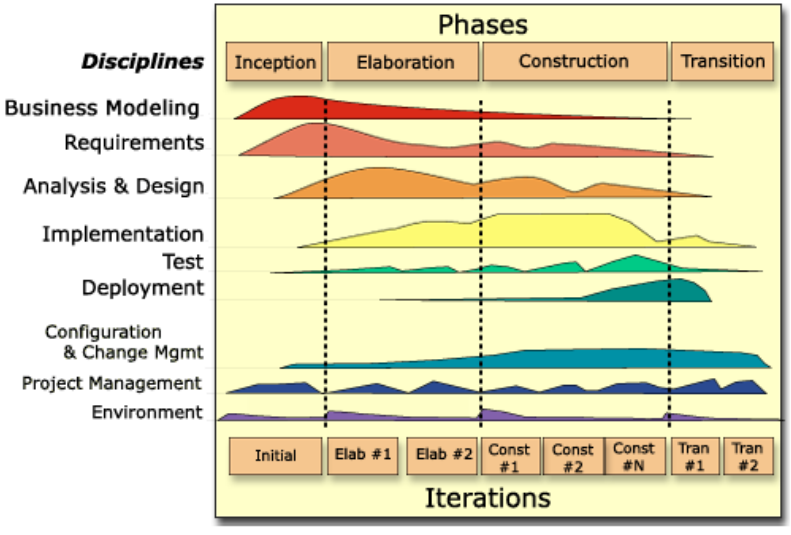
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Software Process: Testing

Completeness important class of faults are suitably targeted

- "important" depends on what you are building
- e.g., if C/C++ is used, beware of memory leaks

Timeliness discover errors as soon as possible

- error in coding revealed at unit testing OK
- error in coding revealed at system integration BAD
- error in coding discovered by final user VERY BAD
- error in the system specifications discovered in system acceptance test CATASTROPHE

Cost effectiveness achieve completeness and timeliness within budget

- on the whole process: do not repeat heavy tasks because of errors

Software Quality Through Testing

- Process visibility: progress must be easily detectable
- This entails that quality goals must be clearly stated and refined
- Goals are measured on software product qualities, which may be:
  - *internal*: only visible to the software developers and designers
    - e.g.: maintainability, reusability, traceability
  - *external*: also visible to final users
    - e.g.: throughput, latency, usability
    - summing up, either dependability or usefulness goals
    - dependability: does it have (functional) faults?
    - usefulness: provided it is dependable, does it have other (typically non-functional) faults?
    - e.g.: bad user interface, software is too slow, etc



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Software Dependability

- Simplest dependability property: correctness
  - all behaviors of the software are as specified
- Reliability: statistical approximation of correctness
  - if not all behaviors are ok, then at least, e.g., 90% of them are
  - often specified w.r.t. a particular usage profile
  - the same program can be more or less reliable depending on how it is used
  - a possible formal definition: percentage of successful operations in a given period $\frac{100|S|}{|S|+|F|}$
    - S is the set of all operations which succeed in the given period

Software Reliability: Other Possibile Definitions

- Availability: reliability w.r.t. failures duration is important
  - may be defined as $100\frac{u}{u+d}$
  - u: software is up and accepting requests
  - d: software is down and not accepting requests
  - typically, $u + d = 1$ day, or 1 week
- MTBF: Mean Time Between Failures
  - may be defined as $\frac{1}{|F|}\sum_{f \in F} |f|$
  - F is the set of all failures in the given period (1 day, 1 week...)
  - for a failure $f \in F$, $|f|$ is the duration, i.e., time required for fixing f
  - more detailed than availability: e.g., it distinguishes from 30 failures of 1 minute and 1 failure of 30 minutes

Software Dependability

- Robustness: correct and reliable only within some defined operational limits
  - if there is a failure only because of a 100x load, the system is however robust
- Safety: nothing bad occurs
  - of course, must be defined w.r.t. some property
  - e.g.: there is never more than one process in the critical section
  - broader sense than what we have defined in model checking
  - essentially, it is any property you can specify, so also liveness and neither liveness nor safety

Terminology

Program or System Under Verification (SUV)

- could also be a part of a "program"
- could also be a system with many processes

Test case A set of inputs, execution conditions, PASS/FAIL criterion

- input is anything the program to be tested can get
  - command-line arguments, files, interrupts, mouse coordinates, sensors...
- execution condition: information on the test execution
  - typically, input timing: whether all input must be provided at the start or not
  - e.g., a sequence of interrupts with given timing
- PASS/FAIL: some way to check
  - e.g.: output must be equal to this expected result

Terminology

Test case specification A formal or informal description of a test case

- "the input is two words" $\rightarrow$ a valid test case will be "goodbye all"

Test suite a set of test cases

Test execution running the test cases on the program

Test obligation a property for test case specifications

- e.g., "all words must be 7 letters long"

Adequacy criteria some property a test suite must fulfill

- e.g., "all test cases must contain at least 30 inputs"
- could also be seen as a set of test obligations
- namely, the adequacy criterion is satisfied if every test obligation is satisfied by at least one test case in the suite



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Terminology

Function Mathematical concept (set of pairs)

Java Function Syntactical function in Java language

- works with all other languages, of course

Unit Smallest unit of work in the program

- typically (but not always) close to single functions or single classes
- here, "unit of work" roughly refers to:
  - the smallest increment by which a software system grows or changes
  - the smallest unit that appears in a project schedule and budget
  - the smallest unit that may reasonably be associated with a suite of test cases (*unit testing*)

Terminology

Independently Testable Feature (ITF) Some functionality of the program which can be isolated from the other functionalities

- not necessary at code level: here, it is testing level
- e.g., a program or a function may be able to both sort and merge files
- however, sorting and merging may be ITF
- granularity depends on the program: from individual functions, to features of an integrated system composed of many programs
- going through individual classes and libraries
- when detected at unit testing, an ITF is usually a function/method or a class, but not only unit testing exists...